UNIT 4 ECOLOGY AND THE ENVIRONMENT

Ecology is the scientific study of the interactions between organisms and their surroundings. It analyses the interactions that living things have with each other and with their non-living, physical environment. In Chapter 14 we look at the components of ecosystems and these interactions. The activities of humans are having many serious effects on the world ecosystems, in particular as a consequence of agricultural practices and industrial pollution. Chapter 15 looks at these activities and considers how humans are damaging the planet.



14 ECOSYSTEMS

An ecosystem is a distinct, self-supporting system of organisms interacting with each other and with a physical environment. An ecosystem can be small, such as a pond, or large, such as a mangrove swamp or a large forest. This chapter looks at a variety of ecosystems and the interactions that happen within them.

LEARNING OBJECTIVES

- Understand the terms ecosystem, habitat, population and community
- Investigate the population size of an organism in two different areas using quadrats

BIOLOGY ONLY

- Understand the term biodiversity
- Investigate the distribution of organisms in their habitats and measure biodiversity usng quadrats

- Understand how biotic and abiotic factors affect the population size and distribution of organisms
- Understand the names given to different trophic levels, including producers, primary, secondary and tertiary consumers, and decomposers
- Understand the concepts of food chains, food webs, pyramids of number, pyramids of biomass and pyramids of energy transfer
- Understand the transfer of substances and energy along a food chain
- Understand why only about 10% of energy is transferred from one trophic level to the next
- Describe the stages in the carbon cycle, including respiration, photosynthesis, decomposition and combustion

BIOLOGY ONLY

- Describe the stages in the nitrogen cycle, including the roles of nitrogen-fixing bacteria, decomposers, nitrifying bacteria and denitrifying bacteria



▲ Figure 14.1 A pond is a small ecosystem.



▲ Figure 14.2 A mangrove swamp is a larger ecosystem.

THE COMPONENTS OF ECOSYSTEMS

Whatever their size, **ecosystems** usually have the same components:

- **producers** – plants which photosynthesise to produce food
- **consumers** – animals that eat plants or other animals
- **decomposers** – organisms that break down dead material and help to recycle nutrients
- the physical environment – all the non-biological components of the ecosystem; for example, the water and soil in a pond or the soil and air in a forest.

KEY POINT

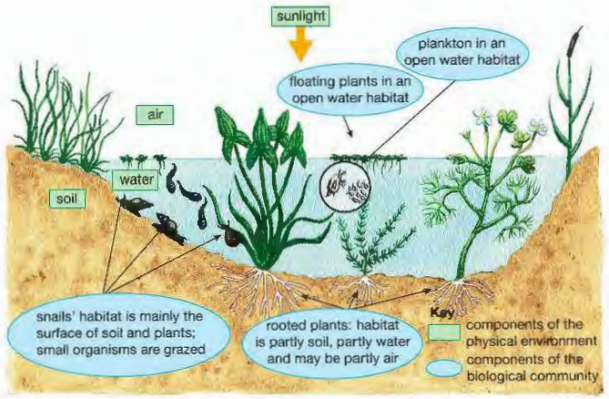
Decomposers are fungi and some species of bacteria. They feed by saprotrophic nutrition (see Chapter 2)

The living components of an ecosystem are called the *biotic* components. The non-living (physical) components are the *abiotic* components (compare these with **biotic** and **abiotic *factors***, below).

An ecosystem contains a variety of habitats. A **habitat** is the place where an organism lives. For example, habitats in a pond ecosystem include the open water, the mud at the bottom of the pond, and the surface water.

All the organisms of a particular species found in an ecosystem at a certain time form the **population** of that species. All the immature frogs (tadpoles) swimming in a pond are a population of tadpoles; all the water lily plants growing in the pond make up a population of water lilies.

The populations of *all* species (animals, plants and other organisms) found in an ecosystem at a particular time form the **community**. Figure 14.3 illustrates the main components of a pond ecosystem.



▲ Figure 14.3 A pond ecosystem

USING QUADRATS TO SAMPLE FROM A HABITAT

HINT

People often mistakenly call quadrats 'quadrants'. A *quadrant* is quite different – it is a quarter of a circle.

When an ecologist wants to know how many organisms there are in a particular habitat, it would not be possible for him to count them all. Instead, he is forced to count a smaller representative part of the population, called a *sample*. Sampling of plants, or animals that do not move much (such as snails), can be done using a sampling square called a **quadrat**. A quadrat is usually made from metal, wood or plastic. The size of quadrat you use depends on the size of the organisms being sampled. For example, to count plants growing on a school field, you could use a quadrat with sides 0.5 or 1 metre in length (Figure 14.4).



▲ Figure 14.4 A student sampling with a quadrat

KEY POINT

Some books (and teachers!) talk about 'throwing' quadrats. The idea is that you stand in the middle of a field and throw the quadrat over your shoulder. This is supposed to be random, but it isn't! The place where the quadrat falls will depend on where you stand, how hard you throw it, etc. It is wrong to use this method.

It is important that sampling in an area is carried out *at random*, to avoid bias. For example, if you were sampling from a school field, but for convenience only placed your quadrats next to a path, this probably wouldn't give you a sample that was representative of the whole field! It would be a *biased* sample.

!

Safety Note: Wash hands after handling the quadrat, plants and soil. Take care if working near a pond or stream or in any areas which may contain animal faeces.

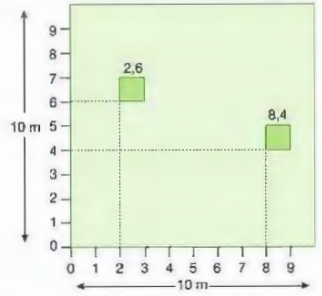
ACTIVITY 1

▼ PRACTICAL: USING QUADRATS TO COMPARE THE SIZE OF A PLANT POPULATION IN TWO AREAS OF A FIELD

One way that you can sample randomly is to place quadrats at coordinates on a numbered grid.

Imagine that there are two areas of a school field (A and B) that seem to contain different numbers of dandelion plants. Area A is more trampled than area B and looks like it contains fewer dandelions. This might lead you to propose the hypothesis: 'The dandelion population in area A is smaller than the dandelion population in area B'.

In area A, two 10-metre tape measures are arranged to form the sides of a square (Figure 14.5).



▲ Figure 14.5 A 10-m² grid with 1-m² quadrats positioned at coordinates 2,6 and 8,4.

A pair of random numbers is generated, using the random number function on a calculator. These numbers are used as coordinates to position the quadrat in the large square.

The numbers of dandelions in the quadrat are counted. The process is then repeated for nine more quadrats.

The tape measures are then moved to area B and the process repeated to sample from ten more quadrats in that part of the field.

The table below shows a set of results for a study like this.

Quadrat number	Number of dandelions in each quadrat in area A	Number of dandelions in each quadrat in area B
1	4	10
2	7	7
3	1	9
4	0	14
5	3	12
6	8	7
7	3	16
8	12	9
9	1	11
10	6	15

Calculate the mean number of dandelions per m^2 in each area. Do the results support the hypothesis that the population numbers are different? How could you improve the reliability of the results?

BIOLOGY ONLY

BIODIVERSITY

The amount of variation shown by species in an ecosystem is called the ecosystem's **biodiversity**. It is a combination of two measurements:

- the number of different species present (known as the *species richness*)
- the relative abundance of each species – their 'evenness' of numbers.

Take the example of the two 'communities' shown in Table 14.1

Table 14.1 Two 'communities' of organisms.

Species	Number of individuals of each species in community 1	Number of individuals of each species in community 2
A	10	1
B	10	1
C	10	1
D	10	1
E	10	46

Both communities contain the same number of species (5) and organisms (50) but community 2 is dominated by one species (E). Community 1 contains an even number of each species, so it has a higher biodiversity.

Some ecosystems such as tropical rainforests have a very high biodiversity (Figure 14.6). Other ecosystems are dominated by one species. This is shown by the pine forest plantations of northern Europe, which are dominated by one species of tree. The trees produce a very dense cover, or 'canopy'. Lack of light severely restricts the growth of other tree species and ground layer plants (Figure 14.7). The pine forest also provides a limited variety of habitats for animals.



▲ Figure 14.6 A tropical rainforest ecosystem contains thousands of different species of plants and animals and has a very high biodiversity.



▲ Figure 14.7 A pine forest has a low biodiversity

EXTENSION WORK

It is possible to calculate a mathematical measure of biodiversity, called a *diversity index*. (See the 'Looking ahead' feature at the end of this chapter.)

Biodiversity is generally a good thing for an ecosystem. Ecosystems with a high biodiversity are often more stable than ones with a low biodiversity. This is because an ecosystem that is dominated by one (or a few) species is more likely to be affected by any sort of ecological disaster. For example, if a new disease arose that wiped out the dominant tree species, this would have an impact on other species that relied on the tree for food and shelter. In a more diverse ecosystem other tree species might supply these resources.

ACTIVITY 2

▼ PRACTICAL: USING QUADRATS TO COMPARE THE BIODIVERSITY OF PLANTS IN TWO HABITATS

The sampling method used in Practical 17 can be modified to compare the biodiversity of plants in two habitats.

Imagine that there are two other areas of the field (C and D) that seem to contain different numbers of various plant species. A hypothesis about this observation might be: 'The diversity of plants in area C is higher than in area D'.

Ten 1 m^2 quadrat samples are taken in each area. The numbers of each plant species present in each quadrat are counted.

! Safety Note:Wash hands after handling the quadrat, plants and soil. Take care if working near a pond or stream or in any areas which may contain animal faeces.

The table below shows a set of results. The student who carried out the investigation was only interested in comparing the broad-leaved plants in the two areas, and did not record the grasses present.

Plant species	Total from 10 quadrats in area C	Total from 10 quadrats in area D
dandelion	69	12
ribwort plantain	78	188
broad-leaved plantain	38	99
daisy	95	22
yarrow	44	12
creeping buttercup	30	5
white clover	49	7
common cat's ear	8	0
groundsel	5	0

Plot the results as two bar charts (total numbers of each species against species name). Use the same axis scales for each bar chart. Describe the biodiversity in the two areas. Do the results support the hypothesis that the two plant communities have a different biodiversity?

END OF BIOLOGY ONLY

INTERACTIONS IN ECOSYSTEMS

The organisms in an ecosystem are continually interacting with each other and with their physical environment. Interactions include the following.

- Feeding among the organisms – the plants, animals and decomposers are continually recycling the same nutrients through the ecosystem.
- Competition among the organisms – animals compete for food, shelter, mates, nesting sites; plants compete for carbon dioxide, mineral ions, light and water.
- Interactions between organisms and the environment – plants absorb mineral ions, carbon dioxide and water from the environment; plants also give off water vapour and oxygen into the environment; animals use materials from the environment to build shelters; the temperature of the environment can affect processes occurring in the organisms; processes occurring in organisms can affect the temperature of the environment (all organisms produce some heat).

REMINDER

Don't forget that plants take in carbon dioxide and give out oxygen only when there is sufficient light for photosynthesis to occur efficiently. When there is little light, plants take in oxygen and give out carbon dioxide. You should be able to explain the reasons for this (see Chapter 10).

BIOTIC AND ABIOTIC FACTORS

There are many factors that influence the numbers and distribution of organisms in an ecosystem. There are two types of factor – *biotic* and *abiotic*.

Biotic factors are biological. Many (but not all) involve feeding relationships. They include:

- availability of food and competition for food resources
- predation
- parasitism
- disease
- presence of pollinating insects
- availability of nest sites.

KEY POINT

A **predator** is an animal that kills and eats another animal. A **parasite** is an organism (animal or plant) that lives in or on another organism (called its host) and gets its nutrition from the host. e.g. mosquitoes are parasites of humans (and other animals), and the human is the host of the mosquito.

Abiotic factors are physical or chemical factors. They include:

- climate, such as light intensity, temperature and water availability
- hours of daylight
- soil conditions, such as clay content, nitrate level, particle size, water content and pH
- other factors specific to a particular habitat, such as salinity (salt content) in an estuary, flow rate in a river, or oxygen concentration in a lake
- pollution.

Clearly *which* factors affect population sizes and distribution of organisms will depend on the *type* of ecosystem. If you take the example of a river, some of the main abiotic factors could be:

- depth of water
- flow rate
- type of material at the bottom of the stream (stones, sand, mud etc.)
- concentration of minerals in the water
- pH
- oxygen concentration
- cloudiness of the water
- presence of any pollution.

The main biotic factors affecting animals in the river will be food supply, either from plants or other animals. But other factors are important too – large fish could not live in a shallow stream!

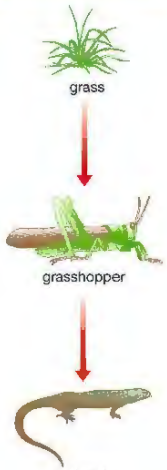
It is impossible to generalise about which factors are the most important. In a heavily polluted river all the organisms could be killed by the pollution, while in a clean river depth and flow rate might have a greater effect on the animals that could live there. The different factors may also affect one another. For example, a faster flow rate could mix the water with air, increasing the amount of dissolved oxygen.

KEY POINT

Large areas of the Earth dominated by a specific type of vegetation are called *biomes*. Temperate grassland and tropical rain forest are two examples of biomes. You could carry out some research to find out about these and other biomes.

The main factor affecting large ecosystems is climate, particularly temperature and rainfall. Climate is the reason why tropical rainforests are restricted to a strip near the equator of the Earth, while pine forests grow in the higher latitudes of the northern hemisphere.

FEEDING RELATIONSHIPS



▲ Figure 14.8 A simple food chain.

KEY POINT

Another way of describing the food chain is that it shows how energy is moved from one organism to another as a result of feeding. The arrows show the direction of energy flow.

The simplest way of showing feeding relationships within an ecosystem is a **food chain** (Figure 14.8).

In any food chain, the arrow (→) means 'is eaten by'. In the food chain illustrated, the grass is the **producer**. It is a plant so it can photosynthesise and produce food materials. The grasshopper is the **primary consumer**. It is an animal which eats the producer and is also a **herbivore**. The lizard is the **secondary consumer**. It eats the primary consumer and is also a **carnivore**. The different stages in a food chain (producer, primary consumer and secondary consumer) are called **trophic levels**.

Many food chains have more than three links in them. Here are two examples of longer food chains:

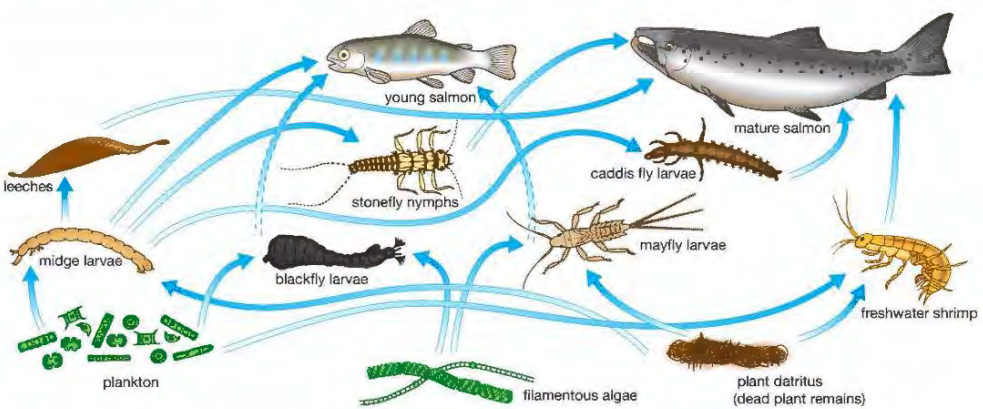
filamentous algae → mayfly nymph → caddis fly larvae → salmon

In this freshwater food chain, the extra link in the chain makes the salmon a **tertiary consumer**.

plankton → crustacean → fish → ringed seal → polar bear

In this marine (sea) food chain, the fifth link makes the polar bear a **quaternary consumer**. Because nothing eats the polar bear, it is also called the *top carnivore*.

Food chains are a convenient way of showing the feeding relationships between a few organisms in an ecosystem, but they oversimplify the situation. The marine food chain above implies that only crustaceans feed on plankton, which is not true. Some whales and other mammals also feed on plankton. For a fuller understanding, you need to consider how the different food chains in an ecosystem relate to each other. Figure 14.9 gives a clearer picture of the feeding relationships involved in a freshwater ecosystem in which salmon are the top carnivores. This is the **food web** of the salmon.



▲ Figure 14.9 The food web of the salmon. As you can see, young salmon have a slightly different diet to mature salmon.

This is still an oversimplification of the true situation, as some feeding relationships are still not shown. It does, however, give some indication of the interrelationships that exist between food chains in an ecosystem. With a little thought, you can predict how changes in the numbers of an organism in one food chain in the food web might affect those in another food chain. For example, if the leech population were to decline through disease, there could be several possible consequences:

- the stonefly nymph population could increase as there would be more midge larvae to feed on
- the stonefly nymph population could decrease as the mature salmon might eat more of them as there would be fewer leeches
- the numbers could remain the same due to a combination of the above.

Although food webs give us more information than food chains, they don't give any information about how many, or what mass of organisms is involved. Neither do they show the role of the decomposers. To see this, we must look at other ways of presenting information about feeding relationships in an ecosystem.

ECOLOGICAL PYRAMIDS

Ecological pyramids are diagrams that represent the relative amounts of organisms at each trophic level in a food chain. There are two main types:

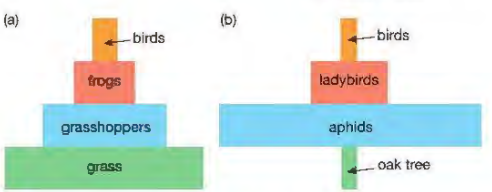
- **pyramids of numbers**, which represent the numbers of organisms in each trophic level in a food chain, irrespective of their mass
- **pyramids of biomass**, which show the total mass of the organisms in each trophic level, irrespective of their numbers.

Consider these two food chains:

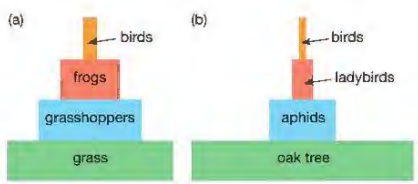
a grass → grasshopper → frog → bird

b oak tree → aphid → ladybird → bird

Figures 14.8 and 14.9 show the pyramids of numbers and biomass for these two food chains.



▲ Figure 14.10 Pyramids of numbers for two food chains



▲ Figure 14.11 Pyramids of biomass for the two food chains

KEY POINT

Biomass is the total mass of organisms. If it refers to living organisms, this is called the *fresh biomass*. More commonly the *dry biomass* is used. This is the mass of plant or animal material after water has been removed, by drying in an oven. Dry biomass is a more reliable measure, since the water content of organisms (especially plants) varies with environmental conditions.

The two pyramids for the 'grass' food chain look the same – the numbers at each trophic level decrease. The *total* biomass also decreases along the food chain – the mass of *all* the grass plants in a large field would be more than that of *all* the grasshoppers which would be more than that of *all* the frogs, and so on.

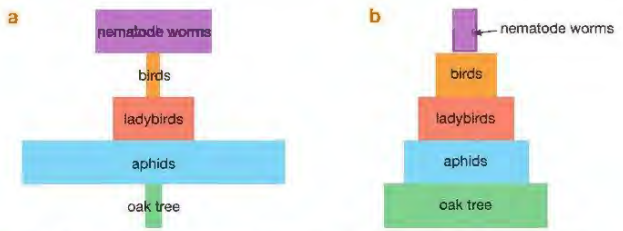
The two pyramids for the 'oak tree' food chain look different because of the size of the oak trees. Each oak tree can support many thousands of aphids, so the numbers *increase* from first to second trophic levels. But each ladybird will need to eat many aphids and each bird will need to eat many ladybirds, so the numbers *decrease* at the third and fourth trophic levels. However, the total biomass *decreases* at each trophic level – the biomass of one oak tree is much

greater than that of the thousands of aphids it supports. The total biomass of all these aphids is greater than that of the ladybirds, which is greater than that of the birds.

Suppose the birds in the second food chain are parasitised by nematode worms (small worms living in the bird's gut). The food chain now becomes:

oak tree → aphid → ladybird → bird → nematode worm

The pyramid of numbers now takes on a very strange appearance (Figure 14.12a) because of the large numbers of parasites on each bird. The pyramid of biomass, however, has a true pyramid shape because the total biomass (Figure 14.12b) of the nematode worms must be less than that of the birds they parasitise.

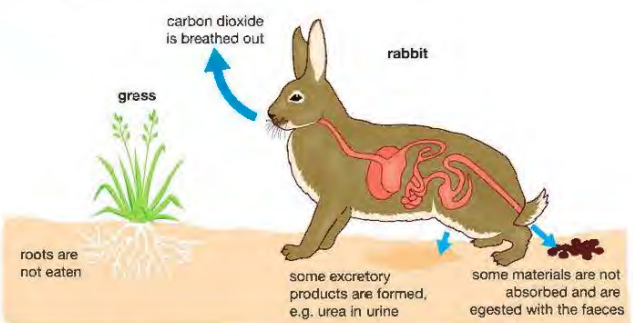


▲ Figure 14.12 (a) A pyramid of numbers and (b) a pyramid of biomass for the parasitised food chain.

WHY ARE DIAGRAMS OF FEEDING RELATIONSHIPS A PYRAMID SHAPE?

The explanation is relatively straightforward (Figure 14.11). When a rabbit eats grass, not all of the materials in the grass plant end up as rabbit! There are losses:

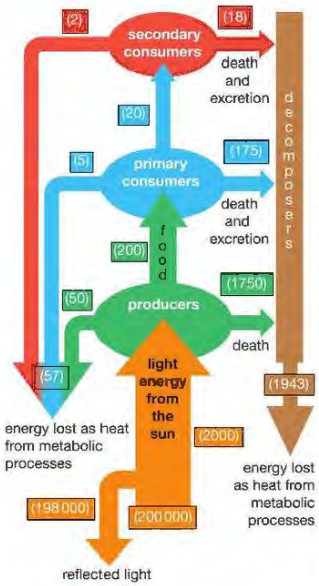
- some parts of the grass are not eaten (the roots for example)
- some parts are not digested and so are not absorbed – even though rabbits have a very efficient digestive system
- some of the materials absorbed form excretory products
- many of the materials are respired to release energy, with the loss of carbon dioxide and water.



▲ Figure 14.13 Not all the grass eaten by a rabbit ends up as rabbit tissue.

In fact, only a small fraction of the materials in the grass ends up in new cells in the rabbit. Similar losses are repeated at each stage in the food chain, so smaller and smaller amounts of biomass are available for growth at successive trophic levels. The shape of pyramids of biomass reflects this.

Feeding is a way of transferring energy between organisms. Another way of modelling ecosystems looks at the energy flow between the various trophic levels.



▲ Figure 14.14 The main ways in which energy is transferred in an ecosystem. The amounts of energy transferred through 1 m² of a grassland ecosystem per year are shown in brackets.

THE FLOW OF ENERGY THROUGH ECOSYSTEMS

This approach focuses less on individual organisms and food chains and rather more on energy transfer between trophic levels (producers, consumers and decomposers) in the whole ecosystem. There are a number of key ideas involved:

- Photosynthesis 'fixes' sunlight energy into chemicals such as glucose and starch.
- Respiration releases energy from organic compounds such as glucose.
- Almost all other biological processes (e.g. muscle contraction, growth, reproduction, excretion, active transport) use the energy released in respiration.
- If the energy released in respiration is used to produce new cells, then the energy remains 'fixed' in molecules in that organism. It can be passed on to the next trophic level through feeding.
- If the energy released in respiration is used for other processes then it will, once used, eventually escape as heat from the organism. Energy is therefore lost from food chains and webs at each trophic level.

Figure 14.14 is an energy flow diagram. It shows the main ways in which energy is transferred in an ecosystem. It also gives the amounts of energy transferred between the trophic levels of this particular (grassland) ecosystem.

As you can see, only about 10% of the energy entering a trophic level is passed on to the next trophic level. This explains why not many food chains have more than five trophic levels. Think of the food chain:

A → B → C → D → E

If we use the idea that only about 10% of the energy entering a trophic level is passed on to the next level, then, of the original 100% reaching A (a producer), 10% passes to B, 1% (10% of 10%) passes to C, 0.1% passes to D and only 0.001% passes to E. There just isn't enough energy left for another trophic level. In certain parts of the world, some marine food chains have six trophic levels because of the huge amount of light energy reaching the surface waters.

CYCLING NUTRIENTS THROUGH ECOSYSTEMS

The chemicals that make up our bodies have all been around before – probably many times! You may have in your body some carbon atoms that were part of the body of Mahatma Gandhi or were in carbon dioxide molecules breathed out by Winston Churchill. This constant recycling of substances is all part of the cycle of life, death and decay.

Microorganisms play a key role in recycling. They break down complex organic molecules in the bodies of dead animals and plants into simpler substances, which they release into the environment.

THE CARBON CYCLE

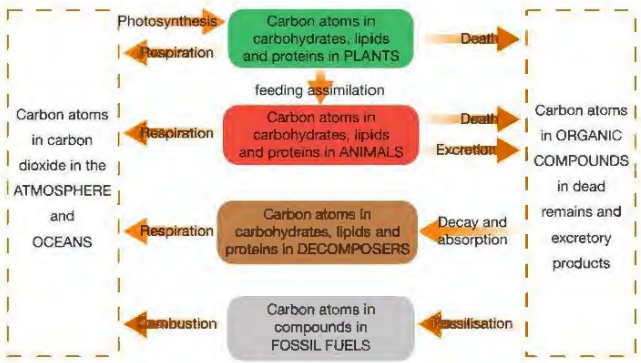
DID YOU KNOW?

What exactly is an organic compound? There is no precise definition! All organic compounds contain carbon, and *most* contain hydrogen. Starch and glucose are organic molecules, but carbon dioxide (CO_2) isn't (it is inorganic).

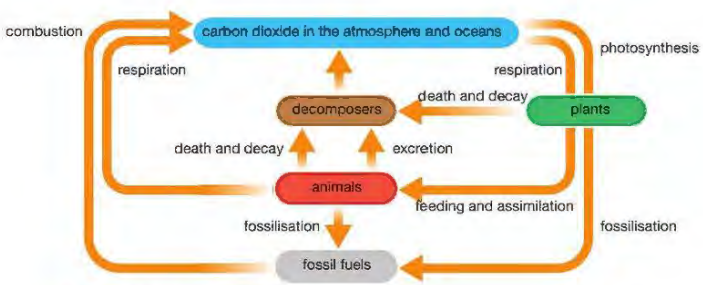
Carbon is a component of all major biological molecules. Carbohydrates, lipids, proteins, DNA, vitamins and many other molecules all contain carbon. The following processes are important in cycling carbon through ecosystems.

- Photosynthesis 'fixes' carbon atoms from carbon dioxide into organic compounds
- Feeding and assimilation pass carbon atoms already in organic compounds along food chains
- Respiration produces inorganic carbon dioxide from organic compounds (mainly carbohydrates) as they are broken down to release energy
- Fossilisation – sometimes living things do not decay fully when they die due to the conditions in the soil (decay is prevented if it is too acidic) and fossil fuels (coal, oil, natural gas and peat) are formed
- Combustion releases carbon dioxide into the atmosphere when fossil fuels are burned.

Figures 14.15 and 14.16 show the role of these processes in the carbon cycle in different ways.



▲ Figure 14.15 The main stages in the carbon cycle



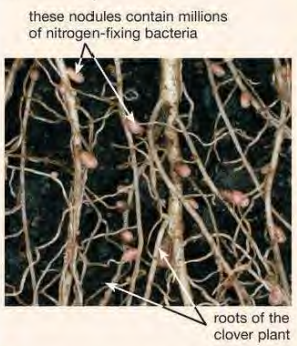
▲ Figure 14.16 A typical illustration of the carbon cycle

BIOLOGY ONLY

THE NITROGEN CYCLE

KEY POINT

Some nitrogen-fixing bacteria are free-living in the soil. Others form associations with the roots of legumes (legumes are plants that produce seeds in a pod, like peas and beans). They form little bumps or 'root nodules' (Figure 14.17).



▲ Figure 14.17 Root nodules on a clover plant.

Instead of entering the soil, the ammonia that the bacteria make by fixing nitrogen is passed to the plant which uses it to make amino acids. In return, the plant provides the bacteria with organic nutrients. This is an example of **mutualism**, where both organisms benefit from the relationship. This nitrogen fixation enriches the soil with nitrates when the plants die and are decomposed.

HINT

To remember if these organic compounds contains nitrogen, check to see if the letter **N** (the symbol for nitrogen) is present:

Protei**N**s, ami**N**o acids and **DNA** all contain nitrogen. Carbohydrates and fats don't – they have no **N**.

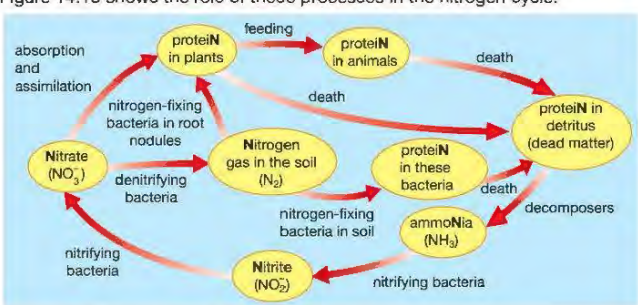
Nitrogen is an element that is present in many biological compounds, including proteins, amino acids, most vitamins, DNA and ATP. Like the carbon cycle, the nitrogen cycle involves feeding, assimilation, death and decay. Photosynthesis and respiration are not directly involved in the nitrogen cycle as these processes fix and release carbon, not nitrogen. The following processes are important in cycling nitrogen through ecosystems.

- Feeding and assimilation pass nitrogen atoms already in organic compounds along food chains.
- **Decomposition** by fungi and bacteria produces ammonia from the nitrogen in compounds like proteins, DNA and vitamins.
- The ammonia is oxidised first to nitrite and then to nitrate by **nitrifying bacteria**. This overall process is called nitrification.
- Plant roots can absorb the nitrates. They are combined with carbohydrates (from photosynthesis) to form amino acids and then proteins, as well as other nitrogen-containing compounds.

This represents the basic nitrogen cycle, but other bacteria carry out processes that affect the amount of nitrate in the soil that is available to plants, as follows:

- **Denitrifying bacteria** use nitrates as an energy source and convert them into nitrogen gas. Denitrification *reduces* the amount of nitrate in the soil.
- Free-living **nitrogen-fixing bacteria** in soil convert nitrogen gas into ammonia. This is used by the bacteria to make amino acids and proteins. When the bacteria die, their proteins decompose, releasing ammonia back to the soil.
- Nitrogen-fixing bacteria in **root nodules** (see margin box) also make ammonia but this is converted by the plant into amino acids and other organic nitrogen compounds. Death and decomposition of the plant returns the nitrogen to the soil as ammonia.

Figure 14.18 shows the role of these processes in the nitrogen cycle.



▲ Figure 14.18 The main stages in the nitrogen cycle

In addition to all the processes described in the biological part of the nitrogen cycle (Figure 14.18), there are other ways that nitrates are formed. Lightning converts nitrogen gas in the air into oxides of nitrogen. These dissolve in rainwater, enter the soil and are converted into nitrates by nitrifying bacteria. Humans also make nitrates industrially from nitrogen gas. These nitrates are mainly used as fertilisers (see Chapter 15.) because they increase the rate of growth of crops.

END OF BIOLOGY ONLY

LOOKING AHEAD – MEASURING BIODIVERSITY

Ecology is a very mathematical branch of biology. An example of this is that biodiversity can be measured by calculating a *diversity index*. One example is called Simpson's Index. One formula for Simpson's Index (D) is:

$$D = 1 - \sum\left(\frac{n}{N}\right)^2$$

In this formula:

N = the total number of individuals of all species

n = the total number of individuals of a particular species.

You calculate $\frac{n}{N}$ for each species, square each of these numbers, and then sum (Σ) all the squares. Subtracting this value from 1 gives you D. Values of D range from 0 to 1, where 0 represents a low diversity and 1 a high diversity.

Take the data you saw in Table 14.1:

Species	Number of individuals of each species in community 1	Number of individuals of each species in community 2
A	10	1
B	10	1
C	10	1
D	10	1
E	10	46

For community 1:

The total number of individuals of all species (N) = 50

For each species the total number of individuals (n) = 10.

So $\frac{n}{N} = \frac{10}{50} = 0.2$ for each species

$$\sum\left(\frac{n}{N}\right)^2 = \left(\frac{10}{50}\right)^2 + \left(\frac{10}{50}\right)^2 + \left(\frac{10}{50}\right)^2 + \left(\frac{10}{50}\right)^2 + \left(\frac{10}{50}\right)^2$$

$$= 0.22 + 0.22 + 0.22 + 0.22 + 0.22$$

$$= 0.04 + 0.04 + 0.04 + 0.04 + 0.04$$

$$= 0.20$$

To find Simpson's Index you subtract this value from 1:

$D = (1 - 0.20) = 0.80$

This value is close to 1.0, showing that community 1 has a high biodiversity.

Can you calculate Simpson's Index for community 2? If you do it carefully you should get the answer $D = 0.152$, showing that community 2 has a lower biodiversity.

EXTENSION WORK

If you are interested, you can do some background research about Simpson's Index. You will find that there are different formulas used for the Index. It doesn't matter which one you use, as long as you are consistent.

CHAPTER QUESTIONS

More questions on ecosystems can be found at the end of Unit 4 on page 221.

SKILLS CRITICAL THINKING

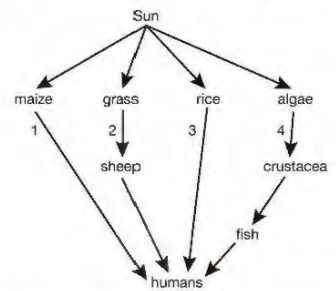
(4) 1 Which of the following terms is defined as 'all the organisms living in a particular place and their interactions with each other and with their environment'?

A habitat **B** population
C community **D** ecosystem

2 Which of the following is a biotic factor?

A light intensity **B** food supply
C pollution **D** temperature

(5) 3 The diagram below shows a food web containing four food chains.



Which food chains are most efficient at using energy from the Sun?

A 1 and 3 **B** 2 and 4 **C** 2 and 3 **D** 3 and 4

BIOLOGY ONLY

(6) 4 In the nitrogen cycle, which of the following conversions is carried out by nitrifying bacteria?

A nitrogen gas to nitrates
B nitrates to nitrogen gas
C ammonia to nitrates
D nitrates to ammonia

END OF BIOLOGY ONLY

(4) 5 a Explain what is meant by the terms habitat, community, environment and population.

(5) b What are the roles of plants, animals and decomposers in an ecosystem?

6 A marine food chain is shown below.

plankton → small crustacean → krill → seal → killer whale

(4) a Which organism is i the producer, ii the secondary consumer?

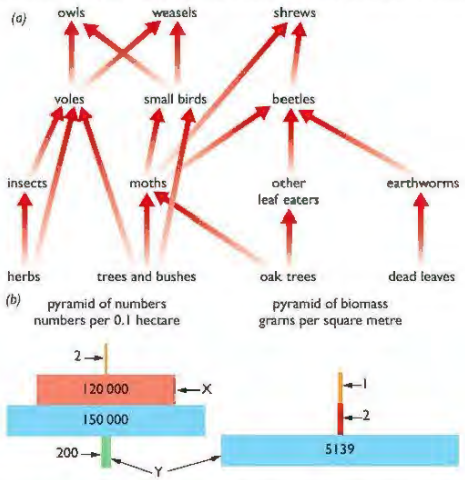
b What term best describes the killer whale?

SKILLS REASONING

(5) c Suggest why five trophic levels are possible in this case, when many food chains only have three or four.

SKILLS INTERPRETATION (6)

7 Part *(a)* of the diagram shows a woodland food web. Part *(b)* shows a pyramid of numbers and a pyramid of biomass for a small part of this wood.



a Write out *two* food chains (from the food web) containing four organisms, both involving moths.

b Name *one* organism in the food web which is both a primary consumer and a secondary consumer.

SKILLS REASONING

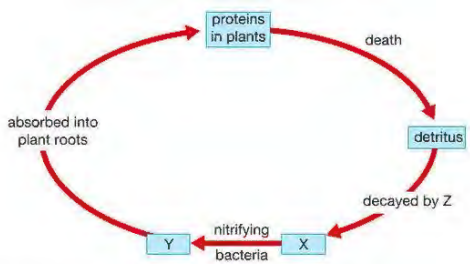
c Suggest how a reduction in the dead leaves may lead to a reduction in the numbers of shrews.

SKILLS ANALYSIS, REASONING

d In part *(b)* of the diagram, explain why level Y is such a different width in the two pyramids.

BIOLOGY ONLY

8 The diagram shows part of the nitrogen cycle.



SKILLS CRITICAL THINKING (6)

a What do X, Y and Z represent?

(5)

b Name the process by which plant roots absorb nitrates.

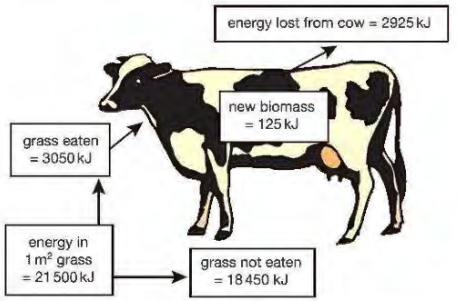
(6)

c What are nitrogen-fixing bacteria?

d Give *two* ways, not shown in the diagram, in which animals can return nitrogen to the soil.

SKILLS PROBLEM SOLVING

9 In a year, 1 m^2 of grass produces 21 500 kJ of energy. The diagram below shows the fate of the energy transferred to a cow feeding on the grass.



(8)

a Calculate the energy efficiency of the cow from the following equation.

$$\text{Energy efficiency} = \frac{\text{energy that ends up as part of cow's biomas}}{\text{energy available}} \times 100$$

SKILLS CRITICAL THINKING (5)

b State two ways that energy is lost from the cow.

SKILLS REASONING (6)

c Suggest what may happen to the 18 450 kJ of energy in the grass that was not eaten by the cow.

SKILLS INTERPRETATION

10 Read the following description of the ecosystem of a mangrove swamp.

Pieces of dead leaves (detritus) from mangrove plants in the water are fed on by a range of crabs, shrimps and worms. These, in turn, are fed on by several species of fish, including young butterfly fish, angelfish, tarpon, snappers and barracuda. Mature snappers and tarpon are caught by fishermen as the fish move out from the swamps to the open seas.

a Use the description to construct a food web of the mangrove swamp ecosystem.

b Write out two food chains, each containing four organisms from this food web. Label each organism in each food chain as producer, primary consumer, secondary consumer or tertiary consumer.

c Decomposers make carbon in the detritus available again to mangrove plants.

SKILLS CRITICAL THINKING

i In what form is this carbon made available to the mangrove plants?

(7)

ii Explain how the decomposers make the carbon available.

END OF BIOLOGY ONLY